



FA report of TP glass chipping issue

JUNGLIM G140HAB01.2

Problem Statement

- General Information

Brand Customer	JUNGLIM	Defect Phenomenon	TP glass chipping
SI Customer		Defect Rate or Q' ty	6pcs
AUO Model	G140HAB01.2	Defect Station	Line Reject
AUO S/N	DF557776CVZZS0100 DF5576R5VHZZS0100 D355A8K9HQZZS0100 DF55A7V3PHZZS0100 DF55A7E6WUZZS0100 D355ABW7MHZZS0100	RMA No.	732612100608
Array/Cell/Beol/Module	L5B/S01	Chip ID	
Week Code	2602	Product Provider	YH
FMA analysis grade		Issue Date	2026/04/16

- Defect Panel S/N Picture

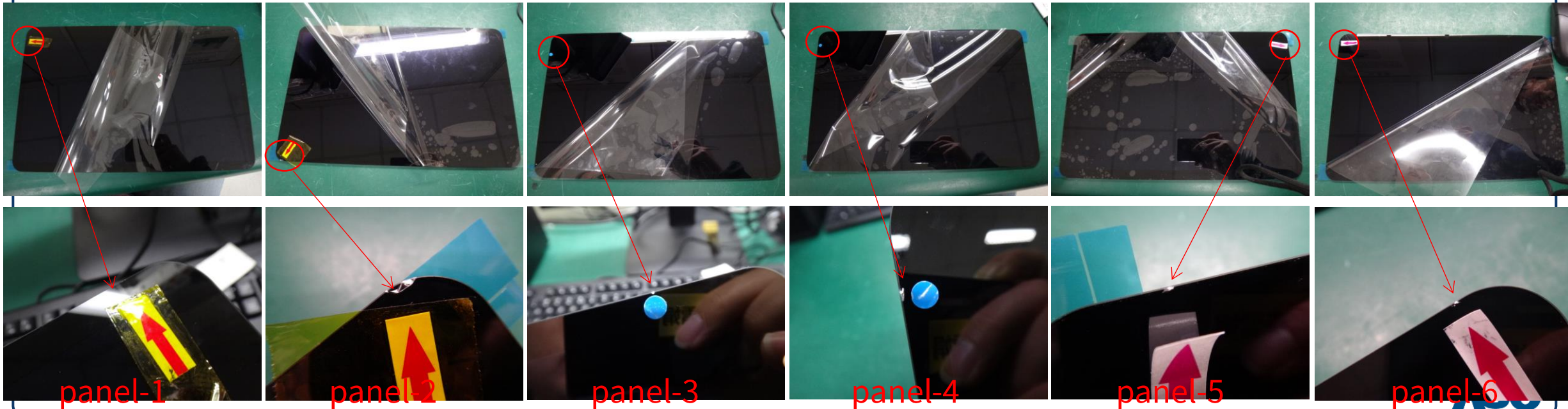


Problem Statement

- Describe the phenomenon

- 1) JUNGLIM identified 6 pieces of G140HAB01.2 panels exhibiting TP glass chipping.
- 2) After verification, the same TP glass chipping phenomenon is also observable at AUO during inspection.

- Defect information



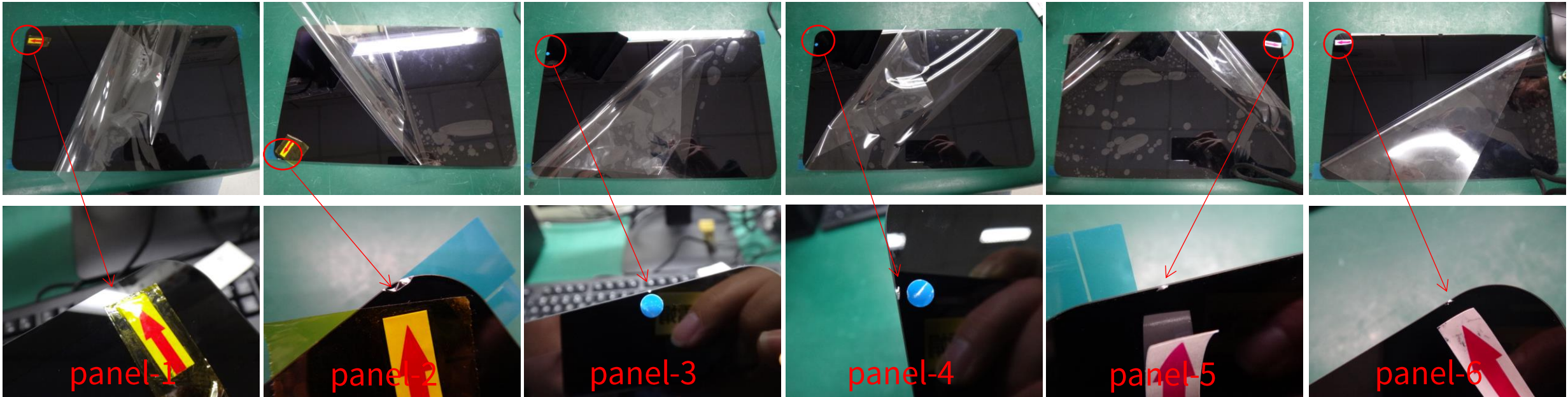
Root Cause Analysis

Step 1 – Phenomenon Confirmation (Visual Inspection)

- According to customer feedback, 6 pcs of panels were reported with TP glass chipping.
- AUO conducted appearance inspection on the returned panels with the following observations:

Panel-1: Fine crack-like chipping observed at the edge of the cover lens.

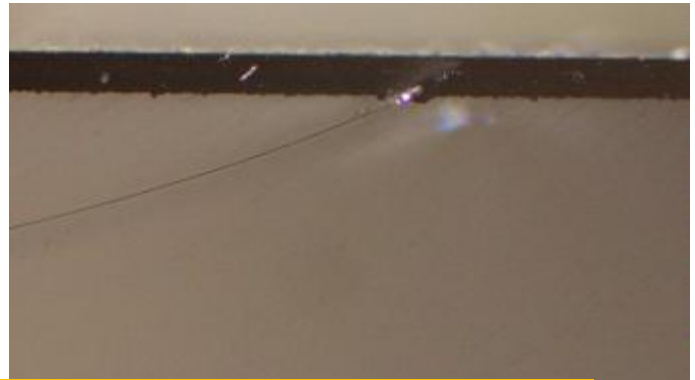
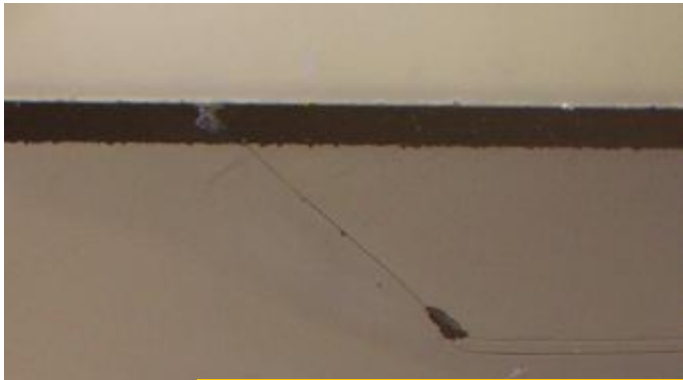
Panel-2 to Panel-6: Minor shell-type chipping observed at the edge of the cover lens.



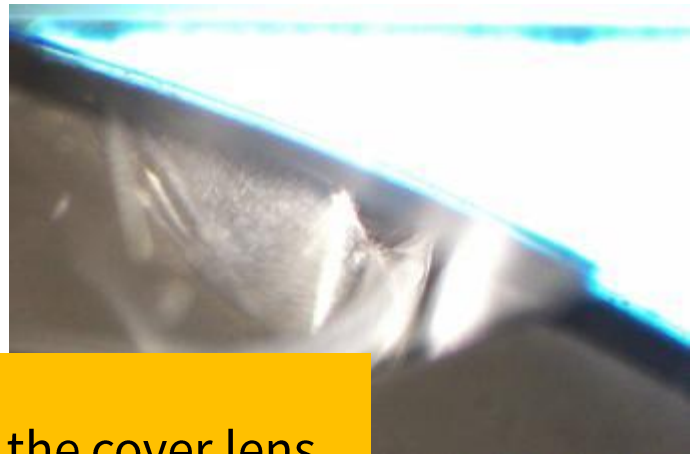
Root Cause Analysis

Step 2 – OM Observation Result

Optical Microscope (OM) inspection was conducted on the defective areas of the returned panels to further evaluate the TP glass chipping condition.



Panel-1 Fine crack-like chipping was observed along the edge of the cover lens.

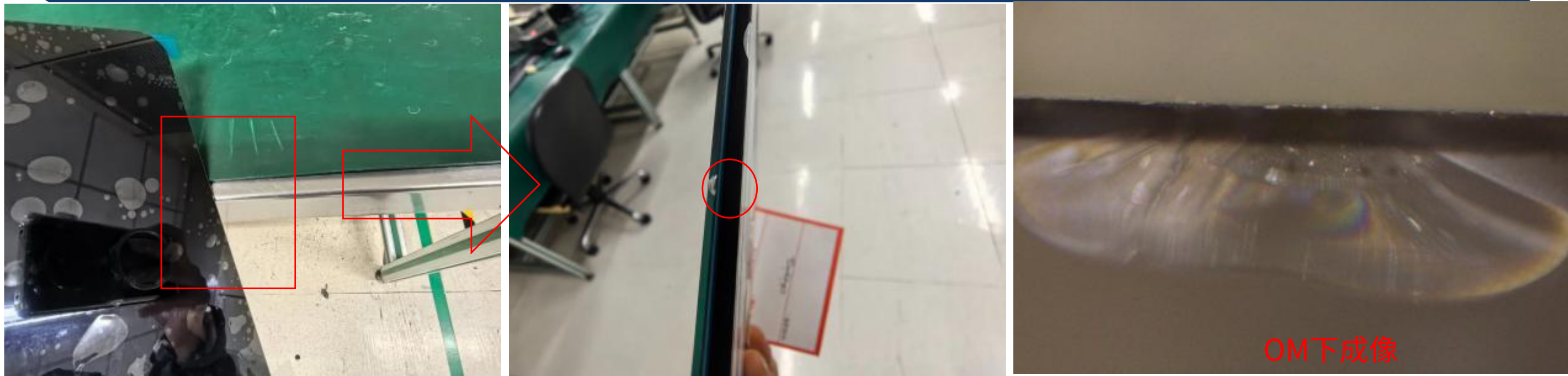


Panel-2 (Representative of Panel-2 to Panel-6)
Shell-type chipping was observed at the edge of the cover lens.

Root Cause Analysis

Step 3 – Simulation Verification

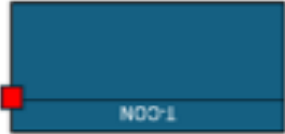
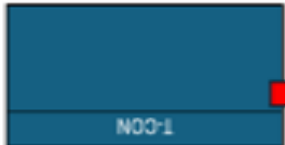
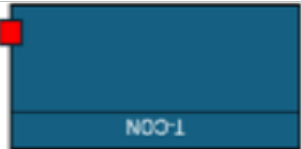
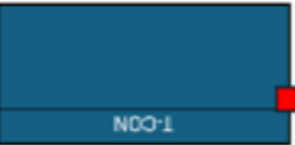
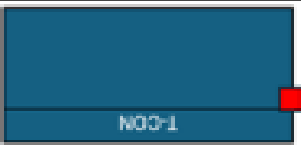
- Edge contact simulation was performed on cover lens.
- Similar shell-type edge chipping can be reproduced.
- OM images of simulated damage are consistent with customer-reported defects. ;



Similar shell-type edge chipping can be reproduced.

G140HAB01.200 TP 裂纹issue

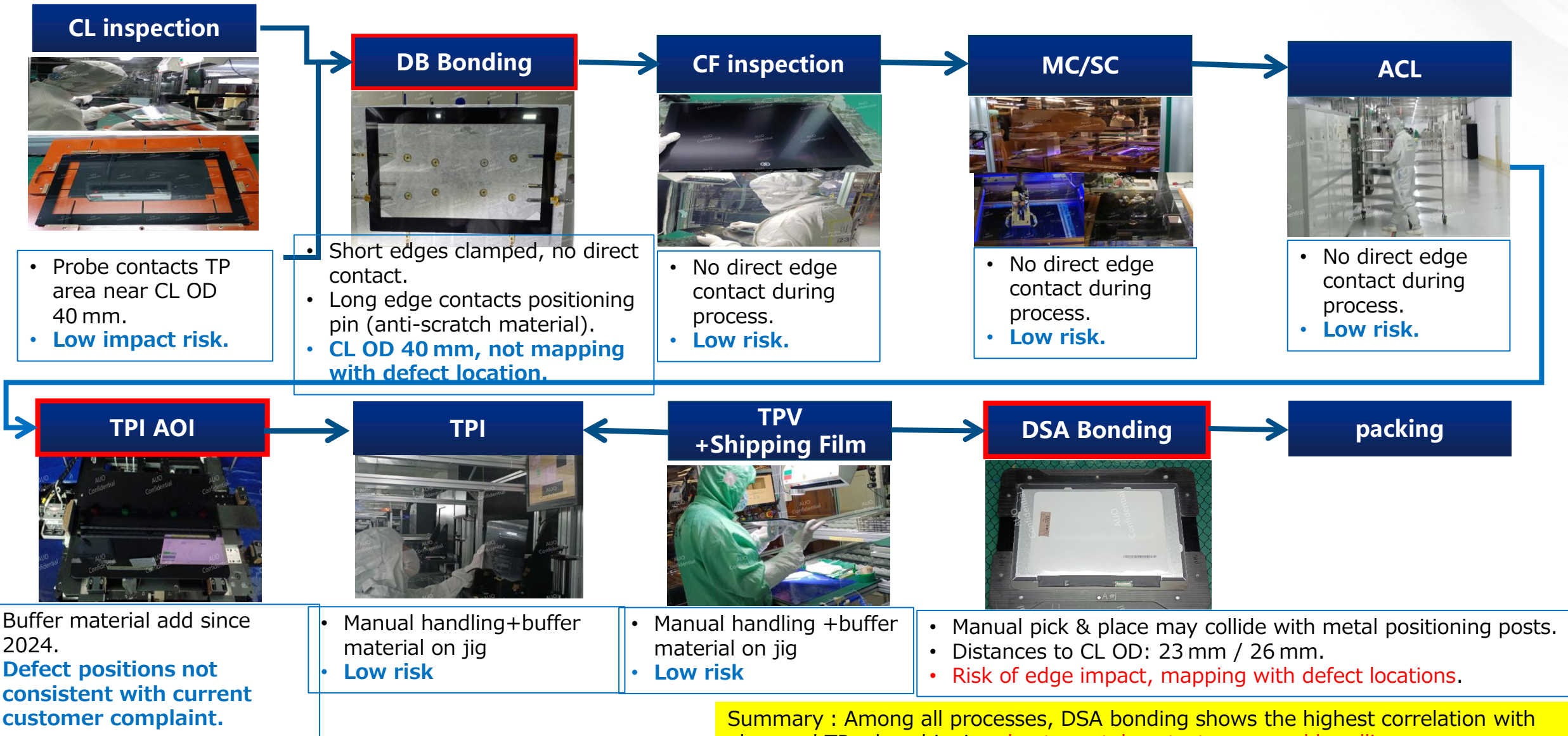
➤ Defect Confirmation

No	Model	SN	W/C	不良图片	位置	距离C/L OD位置
1	G140HAB01.2	DF55A7E6WUZZZS0100	2602			23mm
2	G140HAB01.2	DF5576R5VHZZZS0100	2541			10mm
3	G140HAB01.2	D355A8K9HQZZZS0100	2602			35,mm
4	G140HAB01.2	D355ABW7MHZZZS0100	2602			17mm
5	G140HAB01.2	DF557776CVZZZS0100	2538			25mm
6	G140HAB01.2	DF55A7V3PHZZZS0100	2602			24mm

Summary: All observed defects are edge-related TP cracks, mainly concentrated at the corner / edge region of the panel.

G140HAB01.2 TP glass chipping issue

➤ Full Process Review

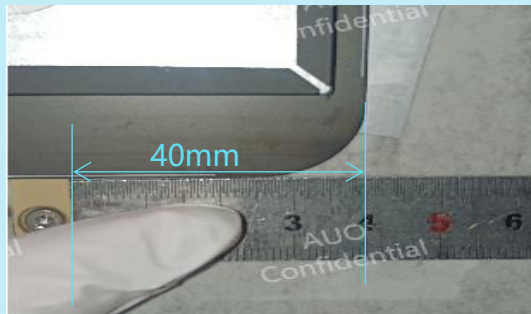


Summary : Among all processes, DSA bonding shows the highest correlation with observed TP edge chipping **due to metal contact + manual handling.**

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➤ 全流程排查：

DB bonding



Summary :

- Short edges fixed by clamp, no direct contact.
 - Long edge contacts positioning pin (anti-scratch material).
 - Distance to C/L OD: 40 mm.
- Not mapping with customer defect position.

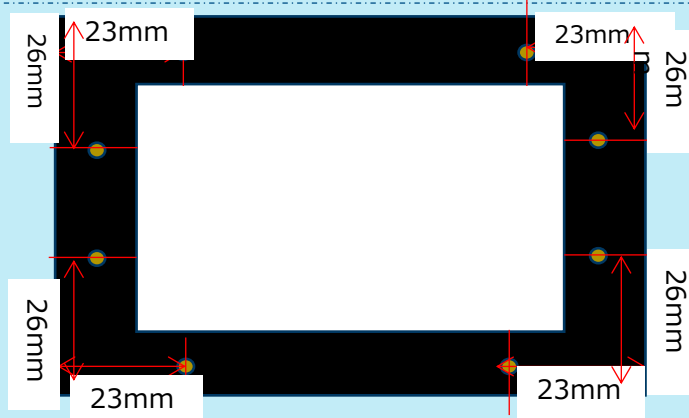
TPI AOI



Summary :

- Two side positions directly contact C/L via pins with pushing motion.
 - Distance to C/L OD: 23 mm.
 - Pins protected with buffer material(since 2024).
- Defect positions not concentrated and different from previous feedback

DSA



Summary :

- Positioning posts may be contacted during manual pick & place.
 - Distance to C/L OD: 23 mm / 26 mm.
- Positioning posts are metal, collision may cause edge chipping.

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➤ Difference Comparison

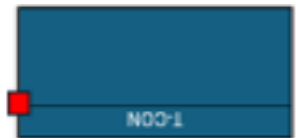
➤ Previous Feedback(2024) :

The phenomenon is fixed at side B, and the distance from the C/L OD edge is approximately 23 mm.

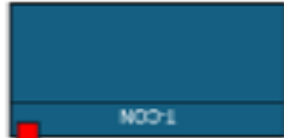


➤ This Time Feedback (2026)

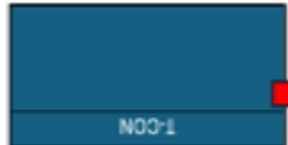
The location is not fixed. Defects appear on three edges, and the distance from the **C/L OD position** is **not fixed**, which is different from the defect location in the previous feedback.



23mm



10mm



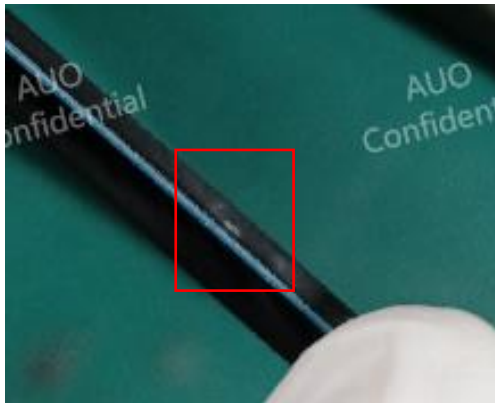
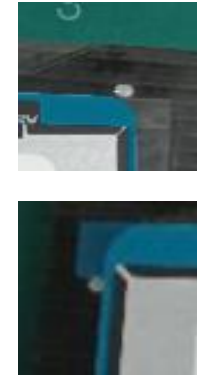
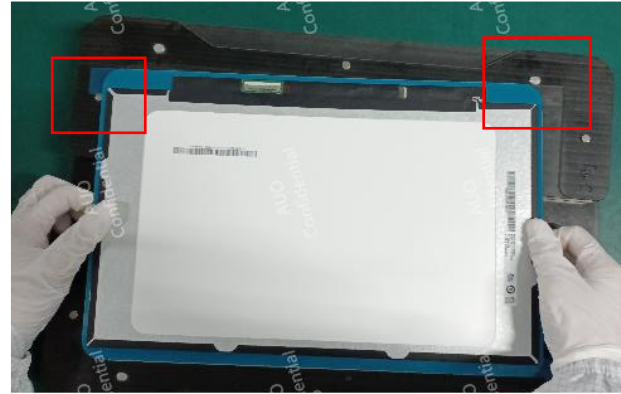
35mm

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➤ Simulation Test

It is initially suspected that edge collision with the positioning post during panel handling (pick-up and placement by operators) leads to impact marks.

The simulation test phenomena are shown as follows :



- Since the overall external contour is a symmetrical structure
- When operators place the panel into the fixture, reverse placement may occur, and during manual adjustment after pick-up, edge collision may occur
- White crack-like changes appear at the impact location

G140HAB01.2 TP glass chipping issue

➤ Improve

➤ Source :

Replace all jig positioning blocks with Bakelite fixtures, in order to avoid edge cracking caused by accidental collision during operator handling. **Estimated completion date: 2026.5.4**



Metal fixtures



Bakelite fixtures

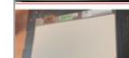


➤ Inspection :

After operators complete DSA tape application, they are required to check whether there are any white impact marks or edge chipping defects around the four edges.

SOP had been implemented on 2026/4/18.



5.3.3	貼附 DSA 膠帶	
	按照金線指引，TP 翻下，將 panel 放入泡棉凹槽中，待膠帶貼 DSA 邊緣後自己；	
	待膠帶貼入後再次放入定位槽中進行定位；	
	貼附 OK 後用手壓住膠帶使其貼緊，膠帶貼合後即翻	
	面邊緣作為定位用；	
	貼附完成後，人員應確認 DSA 貼合位置無偏差，無功不良現象；	

Conclusion and Countermeasure

➤ Conclusion

1. Based on failure analysis, OM observation, simulation verification, and full process review, the TP glass chipping is determined to be caused by external mechanical force applied to the cover lens edge, rather than intrinsic material abnormality.
2. The defect morphology and location are consistent with edge collision during manual handling, especially when the panel edge contacts hard objects.
3. Among all reviewed processes, DSA bonding shows the highest risk correlation, due to manual operation and the use of metal positioning posts near defect-mapping distances.

➤ Countermeasure

Countermeasures implemented:

1. All metal positioning posts have been replaced with Bakelite fixtures to mitigate edge impact risk.
2. An SOP add-on inspection step is introduced to check for white impact marks or edge chipping on all four edges after DSA tape application.
3. These countermeasures aim to eliminate external impact sources and prevent recurrence.



Tap Into The Possibilities

AUO